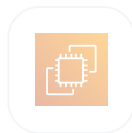


# LionX compute: Amazon ECS

LionX is moving to AWS. First decision to lock in: where its services run.



**ECS Fargate**



ECS on EC2



EKS

The options



The numbers



Our recommendation



**Your GO**

**ADR-0001, proposed**

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# The decision, and what drives it

One choice to make: where LionX compute runs on AWS.

## THE WORKLOAD



### 8 services

long-running, plus cron



### 8 vCPU, 32 GB

per environment



### 5 environments

integration to production

Networking, CI/CD, observability and IAM all hang off this choice.

## WHAT DRIVES THE DECISION

1

### End-2026 cutover.

The platform has to fit the migration runway, not the other way around.

2

### A rewrite either way.

The deployment layer is rewritten regardless. Today's cluster is a delivery vehicle, not a commitment.

3

### Audit surface stays minimal.

Every extra component is more to audit, scan, log and upgrade.

4

### A single workload.

No co-tenants to amortize a cluster's overhead against.

# Three options

Same workload on each, judged on what costs engineer time.

|                   | <br><b>ECS Fargate</b> ✓<br>AWS's container orchestrator, serverless: we ship containers, AWS runs the machines | <br><b>ECS on EC2</b><br>the same orchestrator, on EC2 virtual machines that we own and patch | <br><b>EKS</b> standard to Auto Mode<br>managed Kubernetes: AWS hosts the control plane, we operate the cluster |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KUBERNETES TO RUN | ● none                                                                                                                                                                                           | ● none                                                                                                                                                                           | ● cluster, upgrades, GitOps                                                                                                                                                                        |
| NODE OPERATIONS   | ● AWS's job                                                                                                                                                                                      | ● ours: AMIs, patching, CVEs                                                                                                                                                     | ● ours, or AWS's with Auto Mode                                                                                                                                                                    |
| AUDIT SURFACE     | ● CloudTrail only                                                                                                                                                                                | ● plus OS-level evidence                                                                                                                                                         | ● plus Kubernetes audit logs                                                                                                                                                                       |
| ECOSYSTEM         | ● cron via EventBridge, no Helm                                                                                                                                                                  | ● cron via EventBridge, no Helm                                                                                                                                                  | ● CronJob, Helm, operators                                                                                                                                                                         |
| FIT FOR LIONX     | ● runs it all, nothing unused                                                                                                                                                                    | ● runs everything, ops attached                                                                                                                                                  | ● richer ecosystem, unused                                                                                                                                                                         |

The cheapest bill buys the most ops. Numbers on the next slide, full pros and cons in the ADR.

# The numbers

Yearly, for the LionX footprint. Zurich list prices with a 1-year Savings Plan, ops at a conservative \$1,000 a day.

|                        | <br><b>ECS Fargate</b> ✓ | <br><b>ECS on EC2</b> | <br><b>EKS</b><br>standard to Auto Mode |
|------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| INFRA, PER YEAR        | ~\$4.6K                                                                                                   | ~\$3.1K                                                                                                  | ~\$4.3K to 5.2K                                                                                                            |
| OPS, HOURS PER MONTH   | ~2 to 4                                                                                                   | ~10 to 20                                                                                                | ~15 to 60                                                                                                                  |
| OPS, PER YEAR          | ~\$3K to 6K                                                                                               | ~\$15K to 30K                                                                                            | ~\$23K to 90K                                                                                                              |
| FULLY LOADED, PER YEAR | ● ~\$8K to 11K                                                                                            | ● ~\$18K to 33K                                                                                          | ● ~\$27K to 95K                                                                                                            |

**Fully loaded, Fargate is the cheapest option, not the priciest.** Ops hours decide it.

Infra: eu-central-2 list prices, AWS Pricing API, July 2026. Ops: assumptions built from Encore, CloudAtler, CloudOptimo and Spectro Cloud.

Not measurements. Sourced and detailed per EKS variant in the ADR.

# Decision: ECS Fargate

The smallest operational surface that runs all of LionX.



## ECS Fargate ✓

all eight services and the scheduled tasks, one runtime, zero nodes

### Zero Kubernetes to run

no cluster, no CNI, no GitOps reconciler,  
no upgrade calendar

### One audit surface

CloudTrail end to end, host patching is  
AWS's responsibility

### Nothing blocks landing

no cluster startup on the critical path to  
end 2026

**Accepted trade-offs.** Each schedule is an EventBridge rule plus an IAM role, more Terraform than a CronJob. The ecosystem is narrower than Kubernetes, and the difference would sit unused.

# Next steps

What we ask of you today.



## Questions and objections

here and now, let's settle them in the room



## Ratify today

the ADR moves from proposed to accepted



## PoC

pipeline, IAM boundaries and the cron path, end to end in the new AWS account

**The ask today: your GO on ECS Fargate.**

Any questions? Let's settle them now and call it.